

```

{
    AAA a = new AAA();
    a.disp( );
}

class BBB
{
    public static void disp()
    {
        System.out.println("Hello from
BBB...");
    }

    public static void main(String[]
args)
    {
        BBB b = new BBB();
        b.disp( );
    }
}

```

Figure 6 shows the output of the program *AAA.java*. From the figure, it is clear that the name of the class you use to invoke the Java Virtual Machine (JVM) decides the main method to be called. You can see that the output is different when you execute the commands *java AAA* (the main method of AAA is called) and *java BBB* (the main method of BBB is called). Though surprising, it reminds us that the main method is no way special.

Matrix indexing in MATLAB/Scilab

Imagine students learning their first programming language, be it Python, Java, C, or C++. The first time they come across arrays, I am sure they will have some difficulty adjusting with the indexing starting at 0 (the famous

```

# javac AAA.java
# java AAA
Hello From AAA...
# java BBB
Hello from BBB...
#

```

Figure 6: Output of the Java program *AAA.java*

```

--> x= [111 222 333];

--> x= (0);

Invalid index.

--> x= (1);
ans =

111.

```

Figure 7: Output on the Scilab console

computer scientist Edsger W. Dijkstra has convincingly argued that this should be the case and it would be nice if you could go through his paper). But after a while, the students master the language and become comfortable with array indices starting at 0. Years later, array indices starting at 0 become second nature to them. And then one day they have to learn and use MATLAB/Scilab for some mathematical computations.

All of a sudden, they realise that indexing starts at 1 for MATLAB/Scilab.

You can imagine the number of errors they are going to make within a short span of time. At least, that is what happened to me and it was very frustrating. Figure 7 shows an example for array indexing in the Scilab console. I think the figure clearly illustrates the matter. In addition to MATLAB and Scilab, there are other programming languages like Fortran, Julia, Mathematica, etc, in which arrays start at index 1.

A lot more could have been added to this discussion, but since I don't know whether or not these so called surprises are indeed surprising to others, let us stop for the time being. The selection of surprising features is based on my personal experiences and preferences, so it might be quite possible that the very universally surprising feature Y of programming language X might be missing on this list. But the most important takeaway from this discussion is that while learning a new programming language, if you are really surprised by a particular feature, then it might be the reason why you are learning that language. All the features present in your old programming language will make the learning of the new one easy, but the surprising new features will test you and might even give you a career boost. So, the next time you come across a surprising feature in a programming language, be ready; mastering it might be the break you need. **END** 🐧

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